

Apostol Chapter 1 - Exercise 4

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4. If $(a, b) = 1$, then $(a + b, a - b)$ is either 1 or 2.

Suppose $(a, b) = 1$.

Let $d := (a + b, a - b)$. Then, by definition, $d|(a + b)$ and $d|(a - b)$. Thus, d divides any linear combination of $a + b$ and $a - b$. In particular, it divides their sum and subtraction. Since $(a + b) + (a - b) = 2a$ and $(a + b) - (a - b) = 2b$ we have that $d|(2a)$ and $d|(2b)$. That is, $2a = dq_1$ and $2b = dq_2$, for some integers q_1 and q_2 .

Since $(a, b) = 1$ we can write, $1 = ax + by$, for some integers x, y . Multiplying it by 2 we have

$$\begin{aligned} 2 &= 2ax + 2by \\ &= dq_1x + dq_2y \\ &= d(q_1x + q_2y) \end{aligned}$$

Since, $q_1x + q_2y \in \mathbb{Z}$, we can conclude that $d|2$. That is, if $d = (a + b, a - b)$ then d has to divide 2. Therefore, $(a + b, a - b) \in \{1, 2\}$.